

Cybersecurity Fundamentals Notes

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What is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, systems, and data from cyberattacks, viruses, hackers, and unauthorized access. Its goal is to keep information safe and ensure systems continue working properly.

Why is Cybersecurity Important?

Companies use cybersecurity to:

- Protect important information.
- Prevent hackers from stealing data.
- Keep computers and networks running.
- Protect customer information.
- Avoid financial losses.
- Build customer trust.

The CIA Triad: is the foundation of cybersecurity.

Confidentiality: means that only authorized people can access information.

Example: Only employees with permission can view customer records.

Integrity: means information stays accurate and is not changed without permission.

Example: A customer's bank balance should not be changed by an unauthorized person.

Availability: means systems and data are available whenever users need them.

Example: An online banking website should be accessible 24 hours a day.

Authentication & Authorization

- Authentication: Authentication is the process of verifying a user's identity before they are allowed to access a system.

- Common Authentication Methods:
 - Username and password, Fingerprint, Face recognition, One-Time Password (OTP)
 - Multi-Factor Authentication (MFA)
- Authorization: Authorization is the process of deciding what an authenticated user is allowed to access.

Types of Cyber Threats

- Malware: is harmful software designed to damage or steal information from computers, Examples: Viruses, Worms, Trojans, Spyware, Phishing
- Phishing: is an attack where criminals send fake emails or messages to trick people into revealing passwords or personal information.
- Ransomware: locks files or systems and demands money before access is restored.
- Spyware: secretly collects information from a computer without the user's knowledge.
- Denial-of-Service (DoS) Attack: A DoS attack floods a server or network with traffic until it becomes unavailable to users.

Types of Hackers

- White Hat Hacker: Is an ethical hacker who helps organizations find and fix security weaknesses.
- Black Hat Hacker: Illegally breaks into systems to steal information or cause damage.
- Grey Hat Hacker: May access systems without permission but usually does not intend to cause harm. They often report vulnerabilities afterward.
- Firewalls: is a security system that monitors and controls incoming and outgoing network traffic.
It helps block unauthorized access while allowing trusted communication.
- Antivirus and Anti-malware: software helps detect, prevent, and remove harmful software from computers.
Examples include Microsoft Defender, Bitdefender, Malwarebytes
Keeping antivirus software updated improves protection.

- VPN (Virtual Private Network)

A VPN creates a secure, encrypted connection between a user's device and the internet.

Benefits include:

- Protects personal information
- Secures remote work connections
- Keeps online activity private

- Safe Password Practices

- Good password habits help protect accounts.
- Best practices:
 - Use at least 12 characters.
 - Combine uppercase, lowercase, numbers, and symbols.
 - Do not reuse passwords.
 - Avoid personal information.
 - Change compromised passwords immediately.
 - Use a password manager if possible.

- Social Engineering: is when attackers manipulate people into revealing confidential information,

Examples include:

- Pretending to be IT support.
- Fake phone calls.
- Phishing emails.
- Fake websites.

Always verify requests before sharing sensitive information.

Cybersecurity is an essential part of protecting computers, networks, and information from cyber threats. Understanding the basic concepts, such as the CIA Triad, authentication, authorization, common cyber threats, and security best practices, helps individuals and organizations reduce risks and maintain a secure digital environment. Every user has a role to play in keeping systems and information safe